



2. It is built around quad op-amp LM324 (IC2), transistor 2N3904 (T1), 12V SPDT relay, three 1N4007 diodes, and a few other components.

The heart of the circuit is the quad op-amp IC LM324. This IC has four op-amps, but only three are used in the circuit. The non-inverting input (pin 5) of one op-amp (built around IC1B) is connected to a reference voltage, which is the voltage of the Zener diode, through R2. Unregulated voltage is applied to the inverting input (pin 6) through preset VR1. If the supply voltage is too high, the voltage at the inverting input is more than the reference voltage at the non-inverting input, and the output of the comparator is low.

Similarly, the output voltage of the second op-amp (built around IC1A) is low if the input voltage is low. In this case, the unregulated voltage applied at the inverting input is lower than the reference voltage at the non-inverting input. The outputs of the two comparators are sent to the AND gate consisting of diodes D1 and D2 and pull-up resistor R3. The output is high only if both the inputs are high, that is when the input voltage is within the preset limits.

The output voltage is used to charge capacitor C1. When the voltage on the capacitor is more than the voltage applied to the non-inverting input of the third op-amp (built around IC1C) through VR3 and R5, the output of the op-amp goes high and switches on the relay. The time delay can be adjusted by varying the preset VR3. Diode D3 is used to discharge capacitor C1 when the power is off or beyond set limits.

Red LED2 is used as the power-on indicator and green LED1 lights up when the relay is energised and the input voltage is within limits. The circuit requires an unregulated power supply, which can be ob-

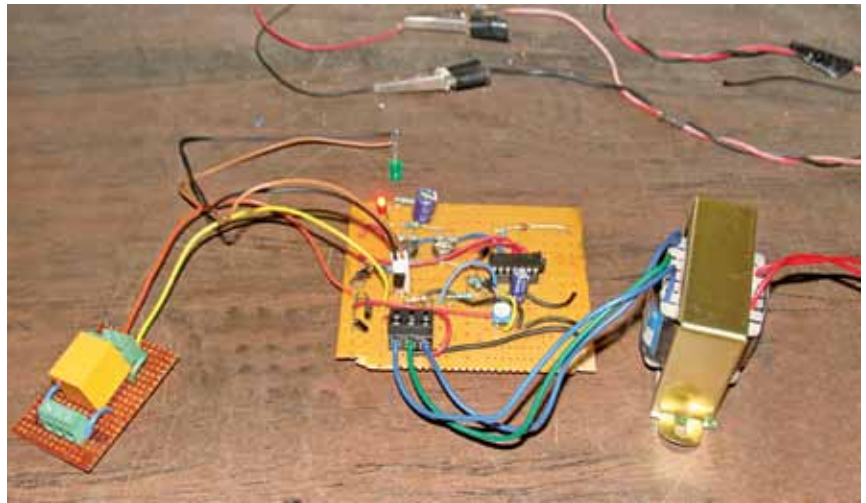


Fig. 3: Author's prototype

tained from a 0-15V, 500mA transformer, rectifier, and a filter (not shown in Fig. 2).

Construction and testing

As the circuit is simple, it can be assembled on a general-purpose PCB or on a breadboard, like the prototype shown in Fig. 1. After assembling the circuit, enclose it in a suitable box. Before operating the circuit, it should be aligned/setup as follows.

Alignment. The circuit can be aligned either by using a variable DC power supply or a Variac. VR1 is used to adjust the upper limit and VR2 to adjust the lower limit of the voltage at which the supply to the load is shut off. Another method of adjusting the circuit is to set the voltage at the wiper of VR1 (high cut) at 1 to 1.5 volts below the Zener voltage and the wiper of VR2 (low cut) at 1 to 1.5 volts below the Zener voltage, with the input at normal voltage.

For adjusting the time delay, a voltage within the set limits should be applied, and when a time equal to the delay required has passed, VR3 should be adjusted until LED1 is on and the relay energises. The resistor through which C1 is

charged is not adjusted; only the preset which applies voltage to the inverting input of the op-amp (built around IC1C) is adjusted, making alignment easier. In this way, the rate at which the capacitor charges is independent of the time setting obtained by adjusting VR3.

The author's prototype made on a general-purpose PCB is shown in Fig. 3. This prototype also includes an unregulated power supply, which is obtained from a 0-15V, 500mA transformer, rectifier, and a filter.

EFY Note. The circuit is not of a voltage stabiliser. If the supply remains either high or low for a long period, a voltage stabiliser should be used. Of course, this circuit can be used along with a stabiliser, and it will cut off power to the load if the voltage is too high or too low and not within the limits even when a stabiliser is used. It will also provide a delay, which is not provided by most stabilisers. **EFY**

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